Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

**COMPUTER SCIENCE DEPARTMENT**

# Software Construction & Development Lab

## Finals Report


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# "Online Quiz Application"

## Introduction

The Online Quiz System is an engaging and dynamic application designed to elevate users' knowledge while offering a highly interactive and educational experience. With an extensive selection of quiz categories and customizable difficulty levels, it ensures that learners of all types can find a suitable challenge tailored to their interests and expertise. Built using Java, the system skillfully incorporates essential programming concepts, including object-oriented principles, exception handling, and input validation to deliver a seamless user experience.
Users can effortlessly register, authenticate, and dive into time-bound quizzes, all while having their progress meticulously tracked through a detailed quiz history. The core features—secure authentication, automated quiz generation, real-time scoring, and robust error handling—ensure the system remains responsive and user-friendly. With a design focused on ease of use and efficiency, this application empowers users to stay engaged, improve their skills, and enjoy the process of learning and testing their knowledge.

## Main Features

1. **User Registration and Authentication:**
   - Registration Number and Password are required to authenticate users before starting the quiz.
   - Validation ensures:

   - Registration number should be exactly 8 digits.
   - Password should be exactly 5 digits.
   - If the input does not meet the criteria, users are prompted to re-enter their details.

2. **User Role:**
   - After authentication, the system assigns the user as a **Regular User**.
   - Users can view their profile and track their quiz history, including the subject and difficulty level of completed quizzes.

3. **Quiz Categories and Difficulty Levels:**
   - Available quiz categories include Math, Science, History, Geography, Literature, and General Knowledge.
   - Users can choose the difficulty level for each quiz: Easy, Medium, or Hard.
   - The system validates the user's choice of category and difficulty level, with error handling in case of invalid input.

4. **Quiz Rules and Instructions:**
   - Before starting the quiz, the user is shown the quiz rules:
     - Time limit (2 minutes for the entire quiz).
     - Answer selection before moving to the next question.
     - No going back after selecting an answer.
   - Users must press Enter to start the quiz.

5. **Quiz Management:**
    - A Quiz class is used to manage the quiz details, such as category, difficulty level, questions, options, and correct answers.
    - The questions and answers are dynamically generated based on the selected subject and difficulty level.
6. **Scoring and Result Display:**
    - The system calculates the score based on the user's answers, displaying the number of correct answers at the end.
    - Incorrect answers are displayed along with the correct answer for reference.
7. **Quiz History:**
    - After completing the quiz, the system adds the quiz to the user's history, which is then displayed.
    - The user can track which quizzes they've taken and the difficulty level of each.
8. **Option Based Answer Selection:**
    - For each question, users are presented with multiple-choice options (A/B/C/D).
    - After the user selects an answer, the system checks if it matches the correct answer.
9. **Reattempt Option**:
    - After completing a quiz, the user is asked if they want to try another quiz. If they answer "no," the system thanks them and ends the session.

## Description of Key Java Concepts (Before Mids)

1. **Java Variables:**
    - Variables in Java are used to store data that can be manipulated during the execution of a program.
    - Variables such as registrationNumber, password, and subject are used to store user input and quiz data.

2. **Data Types:**
    - Data types define the type of data a variable can hold.
    - For example, String holds text, int holds integers, and boolean holds true/false values.
    - The code uses String for text and int for numerical choices.

3. **Loops:**
    - Loops are used to repeat a set of instructions until a certain condition is met.
    - While loops are used to validate user input.

```java
while (regNumber.length() != 8 || !regNumber.matches("\\d+")) {
    System.out.println("\u001B[31mInvalid Registration Number. Please enter exactly 8 digits.\u001B[0m");
    regNumber = sc.nextLine();
}
```

```java
while (password.length() != 5 || !password.matches("\\d+")) {
    System.out.println("\u001B[31mInvalid Password. Please enter exactly 5 digits.\u001B[0m");
    password = sc.nextLine();
}
```

- For loop is used to iterate over quiz questions.

```
for (int i = 0; i < quiz.getQuestions().size(); i++) {
    System.out.println("\n\n" + (i + 1) + ". " + quiz.getQuestions().get(i));
    String[] opts = quiz.getOptions(i);
    for (int j = 0; j < opts.length; j++) {
        System.out.println("    " + (char) ('A' + j) + ". " + opts[j]);
    }
```

4. **Classes:**
   - A class is a blueprint for objects in Java.
   - It contains fields (attributes) and methods (functions) that define the behavior of an object.
   - The code defines classes such as User, RegularUser, and Quiz to structure the program.

```
abstract class User {
    protected String registrationNumber;
    protected String password;

    public User(String registrationNumber, String password) {
        this.registrationNumber = registrationNumber;
        this.password = password;
    }
```

5. **Objects:**
   - Objects are instances of classes.
   - In the code, objects of RegularUser and Quiz are created to represent users and quizzes.

```
RegularUser user = new RegularUser(regNumber, password);
```

```
Quiz quiz = new Quiz(subject, difficultyLevel);
```

6. **Fields:**
   - Fields are variables declared within a class that represent the attributes of an object.
   - In the User class, registrationNumber and password are fields.

```
protected String registrationNumber;
protected String password;
```

```
private List<String> quizHistory = new ArrayList<>();
```

7. **Constructors:**
   - Constructors are special methods used to initialize objects.
   - The constructor of the RegularUser and Quiz classes initializes the object's state.

```
public RegularUser(String registrationNumber, String password) {
    super(registrationNumber, password);
}
```

```java
public Quiz(String category, String difficultyLevel) {
    this.category = category;
    this.difficultyLevel = difficultyLevel;
    this.questions = new ArrayList<>();
    this.options = new ArrayList<>();
    this.correctAnswers = new ArrayList<>();
}
```

8. **Accessors and Mutators (Getters and Setters):**
   - Accessor methods (getters) are used to retrieve the value of fields.

```java
public String getCategory() {
    return category;
}

public String getDifficultyLevel() {
    return difficultyLevel;
}
```

   - Mutator methods (setters) are used to modify them.

```java
public void addQuestion(String question, String[] options, String correctAnswer) {
    questions.add(question);
    this.options.add(options);
    correctAnswers.add(correctAnswer);
}
```

9. **Parameters:**
   - Parameters are values passed to methods to provide them with necessary input.
   - In this code, methods like startQuiz accept parameters such as quiz, user, and sc to process the quiz.

```java
public void addQuizToHistory(String quizName) {
    quizHistory.add(quizName);
}
```

```java
private static void startQuiz(Quiz quiz, RegularUser user, Scanner sc) {
    int score = 0;
    List<String> incorrectAnswers = new ArrayList<>();
```

10. **Assignment:**
   - Assignment involves assigning a value to a variable.
   - Variables such as subject, difficultyLevel, and score are assigned values.

```java
// Subject Selection
boolean validSubject = false;
String subject = "";
while (!validSubject) {
```

```
// Difficulty Level Selection
boolean validDifficulty = false;
String difficultyLevel = "";
while (!validDifficulty) {
```

```
int score = 0;
```

**11. Conditional Statements:**

- Conditional statements like if, else, and switch are used to execute code based on conditions.
- For instance, the code uses an if statement to validate input and a switch statement to select a difficulty level.

```
if (subjectChoice >= 1 && subjectChoice <= 6) {
    subject = subjects[subjectChoice - 1];
    validSubject = true;
```

```
switch (difficultyChoice) {
case 1:
    difficultyLevel = "Easy";
    System.out.println(GREEN + "You selected Easy!" + RESET);
    validDifficulty = true;
    break;
case 2:
    difficultyLevel = "Medium";
    System.out.println(YELLOW + "You selected Medium!" + RESET);
    validDifficulty = true;
    break;
case 3:
    difficultyLevel = "Hard";
    System.out.println(RED + "You selected Hard!" + RESET);
    validDifficulty = true;
    break;
```

**12. Scanner:**

- The Scanner class is used for capturing user input from the console.
- It's employed to read values like registration numbers, passwords, and subject choices.

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    try {
        // User Authentication
        System.out.println("Welcome to the Online Quiz System");

        System.out.print("Enter Registration Number (8 digits): ");
        String regNumber = sc.nextLine();
```

```
System.out.print("Enter Password (5 digits): ");
String password = sc.nextLine();
```

### 13. Polymorphism:

- Polymorphism allows methods to have different behaviors based on the object that calls them.
- In this code, displayRole() is polymorphic because it behaves differently depending on whether the object is a User or RegularUser.

```
public abstract void displayRole();
```

```
@Override
public void displayRole() {
    System.out.println("Role: Regular User");
}
```

### 14. Inheritance:

- Inheritance enables a class to inherit methods and properties from another class.
- The RegularUser class inherits from the User class, allowing it to use the registrationNumber and password fields.

```
class RegularUser extends User {
    private List<String> quizHistory = new ArrayList<>();

    public RegularUser(String registrationNumber, String password) {
        super(registrationNumber, password);
    }
```

### 15. Abstraction:

- Abstraction hides implementation details and shows only essential features.
- The User class is abstract, which means it cannot be instantiated directly and provides a base for other classes like RegularUser.

```
abstract class User {
    protected String registrationNumber;
    protected String password;
```

```
public abstract void displayRole();
```

### 16. Encapsulation:

- Encapsulation is the concept of wrapping data and methods into a single unit.
- This code uses private fields and public methods to ensure proper access and modification.

```
protected String registrationNumber;
protected String password;
```

## Description of Key Java Concepts (Finals)

### 17. Loose Coupling:

- Loose coupling means that components are independent of each other.
- The code structure allows different components (like RegularUser and Quiz) to interact without tightly depending on each other.

```
Quiz quiz = new Quiz(subject, difficultyLevel); // Loose coupling between Quiz and RegularUser
generateQuestions(quiz);
```

### 18. High Cohesion:

- High cohesion is maintained by ensuring that each class has a single responsibility.
- The Quiz class manages only quiz-related functionality, and the User class handles user-related actions.

```
class Quiz {
    private String category;
    private String difficultyLevel;
    private List<String> questions;
    private List<String[]> options;
    private List<String> correctAnswers;
```

### 19. Refactoring:

- The code can be refactored to improve readability and maintainability.
- This code has sections where the flow could be simplified or improved, but it's already modular and clean.

```
private static void generateQuestions(Quiz quiz) {
    String subject = quiz.getCategory();
    try {
        switch (subject) {
        case "Math":
            if (quiz.getDifficultyLevel().equals("Easy")) {
                quiz.addQuestion("What is 5 + 3?", new String[]{"6", "7", "8", "9"}, "C");
                quiz.addQuestion("Solve: 10 - 2.", new String[]{"8", "7", "3", "9"}, "A");
                quiz.addQuestion("What is 4 multiplied by 2?", new String[]{"6", "8", "10", "12"}, "B");
                quiz.addQuestion("Divide 20 by 5.", new String[]{"2", "3", "4", "5"}, "C");
                quiz.addQuestion("What is the square of 3?", new String[]{"6", "7", "8", "9"}, "D");
            } else if (quiz.getDifficultyLevel().equals("Medium")) {
                quiz.addQuestion("What is 15% of 200?", new String[]{"25", "30", "35", "40"}, "B");
                quiz.addQuestion("Solve: 12 × (4 + 2).", new String[]{"60", "72", "48", "24"}, "B");
                quiz.addQuestion("What is the square root of 81?", new String[]{"7", "8", "9", "10"}, "C");
                quiz.addQuestion("If x = 7, find the value of 3x + 2.", new String[]{"23", "24", "25", "26"}, "A");
                quiz.addQuestion("Simplify: (5/4) + (3/4).", new String[]{"1", "2", "3", "4"}, "B");
            } else if (quiz.getDifficultyLevel().equals("Hard")) {
                quiz.addQuestion("What is 25 × 24?", new String[]{"600", "575", "625", "480"}, "A");
                quiz.addQuestion("Solve: (18 × 3) - (12 ÷ 2).", new String[]{"48", "45", "51", "43"}, "A");
                quiz.addQuestion("Find the value of x in 3x + 5 = 20.", new String[]{"5", "7", "6", "8"}, "A");
                quiz.addQuestion("What is the value of 10³?", new String[]{"100", "1000", "10000", "100000"}, "B");
                quiz.addQuestion("Simplify: (7 × 3) - (2 × 4).", new String[]{"13", "17", "19", "20"}, "A");
            }
            break;
```

### 20. Exception Handling:

- Exception handling is implemented to catch invalid inputs, such as incorrect registration number formats or out-of-bound subject choices.

- For example, try-catch blocks are used around registration number and password validation to provide meaningful error messages.

```java
try {
    // User Authentication
```

```java
        }
    } catch (Exception e) {
        System.out.println("An error occurred: " + e.getMessage());
        e.printStackTrace();
    } finally {
        sc.close();
    }
}
```

```java
try {
    for (int i = 0; i < quiz.getQuestions().size(); i++) {
        System.out.println("\n\n" + (i + 1) + ". " + quiz.getQuestions().get(i));

        // Defensive programming for options retrieval
        if (quiz.getOptions(i) == null || quiz.getOptions(i).length == 0) {
            System.out.println("Error: No options available for this question.");
            continue; // Skip this question
        }

        String[] opts = quiz.getOptions(i);
        for (int j = 0; j < opts.length; j++) {
            System.out.println("    " + (char) ('A' + j) + ". " + opts[j]);
        }

        System.out.print("Your Answer (A/B/C/D): ");
        String answer = sc.nextLine().toUpperCase();

        // Validate user input
        if (!answer.matches("[A-D]")) {
            System.out.println("Invalid input. Please enter A, B, C, or D.");
            i--; // Repeat the question
            continue;
        }

        String correctAnswer = quiz.getCorrectAnswer(i);
        if (correctAnswer == null) {
            System.out.println("Error: No correct answer found for this question.");
            continue; // Skip this question
        }

        if (answer.equals(correctAnswer)) {
            score++;
        } else {
            incorrectAnswers.add((i + 1) + ". " + quiz.getQuestions().get(i));
            incorrectAnswers.add("Correct Answer: " + correctAnswer);
        }
    }
} catch (IndexOutOfBoundsException e) {
    System.out.println("Error: Question or answer index is out of bounds. " + e.getMessage());
} catch (Exception e) {
    System.out.println("An unexpected error occurred during the quiz: " + e.getMessage());
}
```

**21. Dynamic Polymorphism:**

- Dynamic polymorphism occurs when a method call is resolved at runtime.
- It is demonstrated in the code through method overriding (e.g., displayRole() in the User class).

```java
user.displayRole();
```

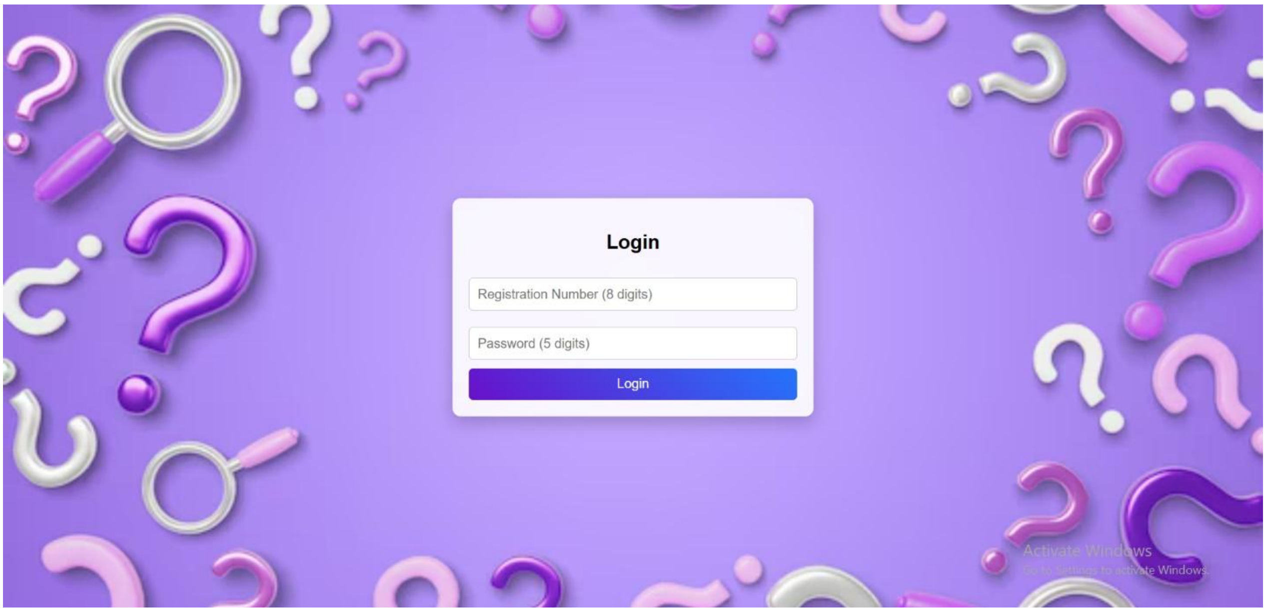
## Output

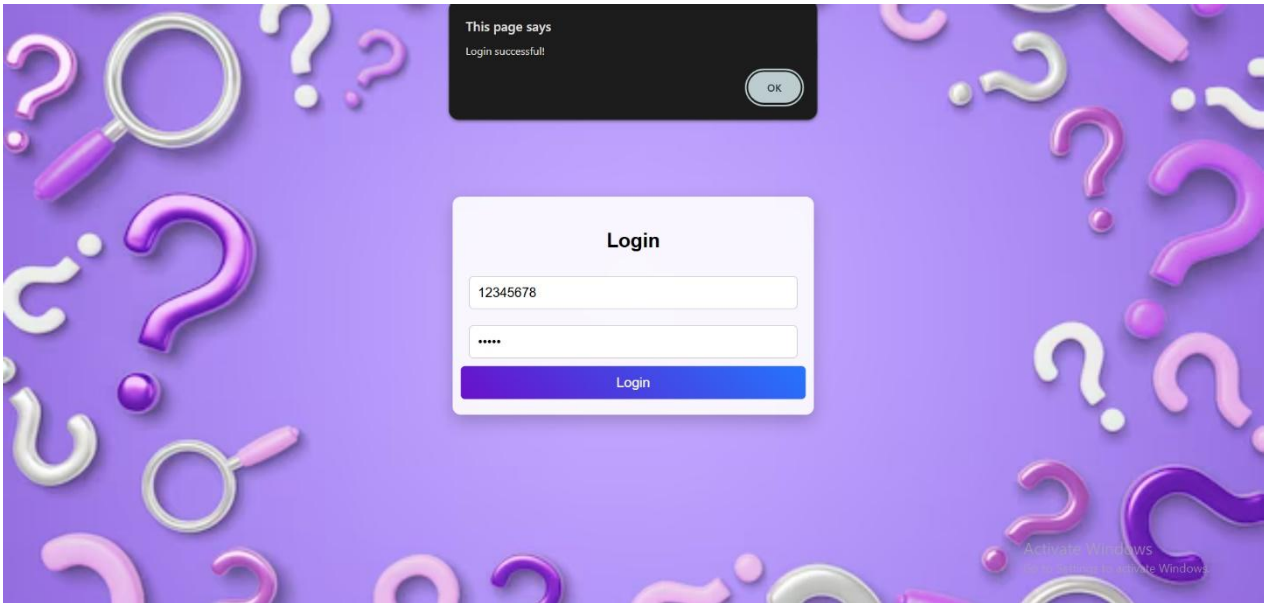
```
OnlineQuiz [Java Application] C:\Users\The Laptop Hut\.p2\pool\plugins\org.eclipse.justj.ope
Welcome to the Online Quiz System
Enter Registration Number (8 digits): 123456789
Invalid Registration Number. Please enter exactly 8 digits.
12345678
Enter Password (5 digits): 123456
Invalid Password. Please enter exactly 5 digits.
12345
Role: Regular User

╔══════════════════════════════╗
        Available Subjects
╚══════════════════════════════╝
  1. Math
  2. Science
  3. History
  4. Geography
  5. Literature
  6. General Knowledge
Choose a subject (1-6): 1

╔══════════════════════════════════╗
      Choose Difficulty Level

╚══════════════════════════════════╝
  1. Easy   - Take it light and simple!
  2. Medium - A balanced challenge awaits.
  3. Hard   - Only for the brave!
Enter Difficulty Level (1-3): 1
You selected Easy!
========================================
****     QUIZ RULES     ****

========================================
1. You only have 2 minutes to answer all questions.
2. You have to select an option before you can go to the next question.
3. There's no going back after selecting an option.
4. At the end of the quiz, your score will be displayed.
Good luck, and give your best shot!

========================================

Press Enter to start the quiz...


Quiz Starting...


1. What is 5 + 3?
   A. 6
   B. 7
   C. 8
   D. 9
Your Answer (A/B/C/D): c
```

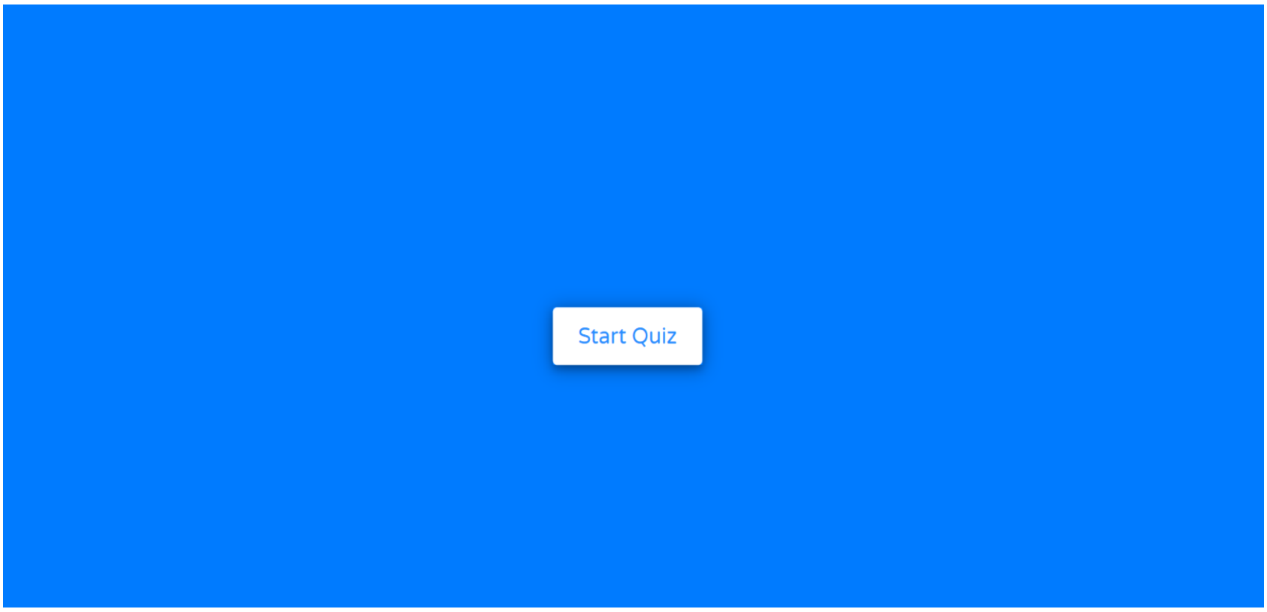
```
2. Solve: 10 - 2.
   A. 8
   B. 7
   C. 3
   D. 9
Your Answer (A/B/C/D): a


3. What is 4 multiplied by 2?
   A. 6
   B. 8
   C. 10
   D. 12
Your Answer (A/B/C/D): b


4. Divide 20 by 5.
   A. 2
   B. 3
   C. 4
   D. 5
Your Answer (A/B/C/D): c


5. What is the square of 3?
   A. 6
   B. 7
   C. 8
   D. 9
Your Answer (A/B/C/D): d

Quiz Completed!
Your Score: 5/5
Quiz History: [Math Quiz (Easy)]
All answers were correct!
Do you want to continue to another quiz? (yes/no): yes

╔══════════════════════════════════╗
        Available Subjects
╚══════════════════════════════════╝
  1. 📘 Math
  2. 🔬 Science
  3. 📜 History
  4. 🌍 Geography
  5. 📖 Literature
  6. ❓ General Knowledge
Choose a subject (1-6): 2

╔══════════════════════════════════════╗
      ⚡ Choose Difficulty Level ⚡

╚══════════════════════════════════════╝
  1. ✨ Easy   - Take it light and simple!
  2. ⚡ Medium - A balanced challenge awaits.
  3. 🔥 Hard   - Only for the brave!
Enter Difficulty Level (1-3): 2
You selected Medium!

========================================
****    QUIZ RULES    ****

========================================
🚀 1. You only have 2 minutes to answer all questions.
⏱ 2. You have to select an option before you can go to the next question.
```

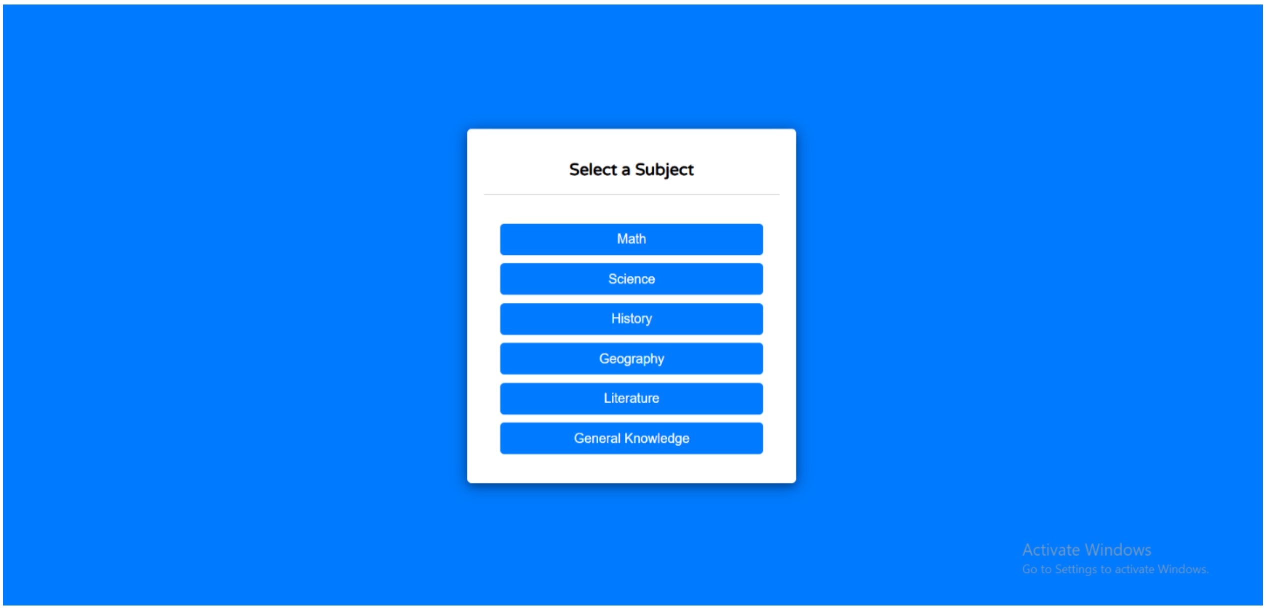
```
🤔 3. There's no going back after selecting an option.
🏆 4. At the end of the quiz, your score will be displayed.
⚡ Good luck, and give your best shot! ⚡

========================================

Press Enter to start the quiz...


Quiz Starting...


1. What is the atomic number of carbon?
   A. 6
   B. 12
   C. 8
   D. 10
Your Answer (A/B/C/D): a


2. Which organelle is responsible for energy production in cells?
   A. Nucleus
   B. Mitochondria
   C. Endoplasmic Reticulum
   D. Golgi Apparatus
Your Answer (A/B/C/D): b


3. What is the chemical formula for methane?
   A. CO2
   B. H2O
   C. CH4
   D. CO
Your Answer (A/B/C/D): d

4. Which gas is responsible for global warming?
   A. Oxygen
   B. Nitrogen
   C. Carbon Dioxide
   D. Hydrogen
Your Answer (A/B/C/D): d


5. What is the boiling point of nitrogen?
   A. -180°C
   B. -196°C
   C. -150°C
   D. -210°C
Your Answer (A/B/C/D): c

Quiz Completed!
Your Score: 2/5
Quiz History: [Math Quiz (Easy), Science Quiz (Medium)]

Incorrect Answers:
3. What is the chemical formula for methane?
Correct Answer: C
4. Which gas is responsible for global warming?
Correct Answer: C
5. What is the boiling point of nitrogen?
Correct Answer: B

Do you want to continue to another quiz? (yes/no): no
Thank you for taking the quiz! Goodbye.
```

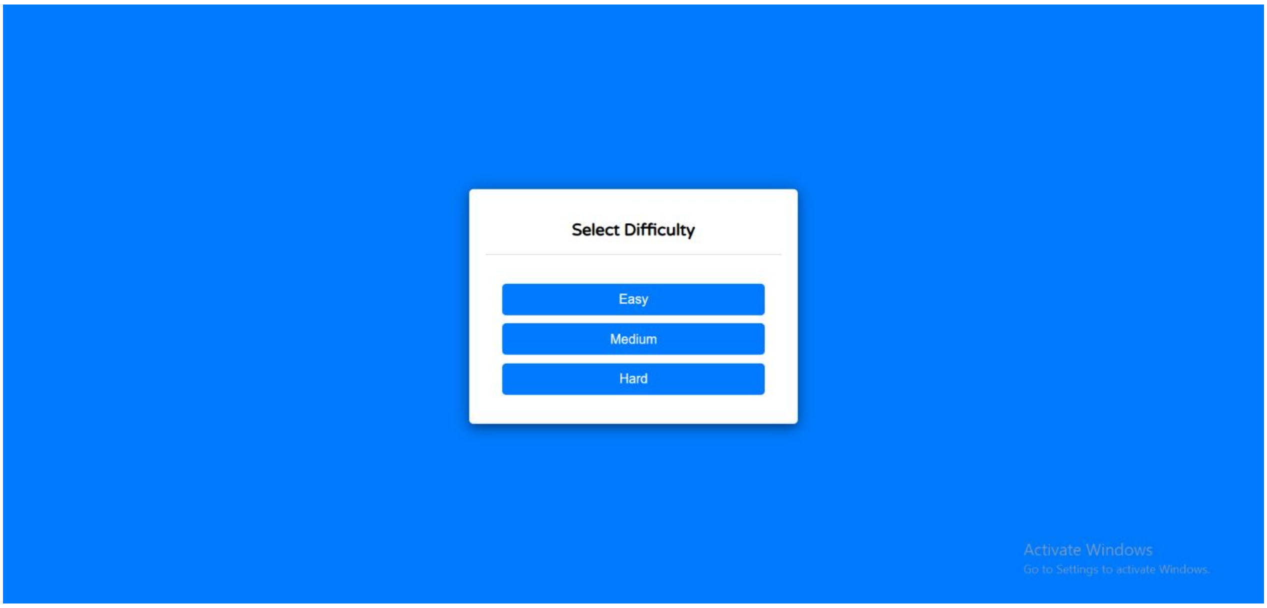
## Graphical User Interface

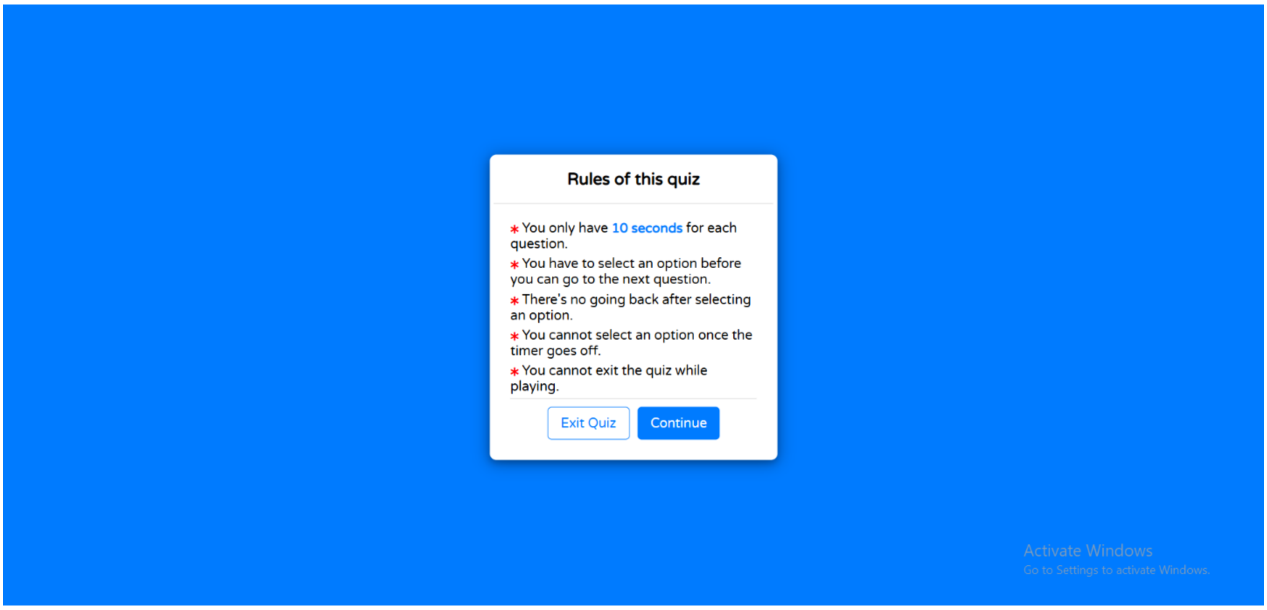


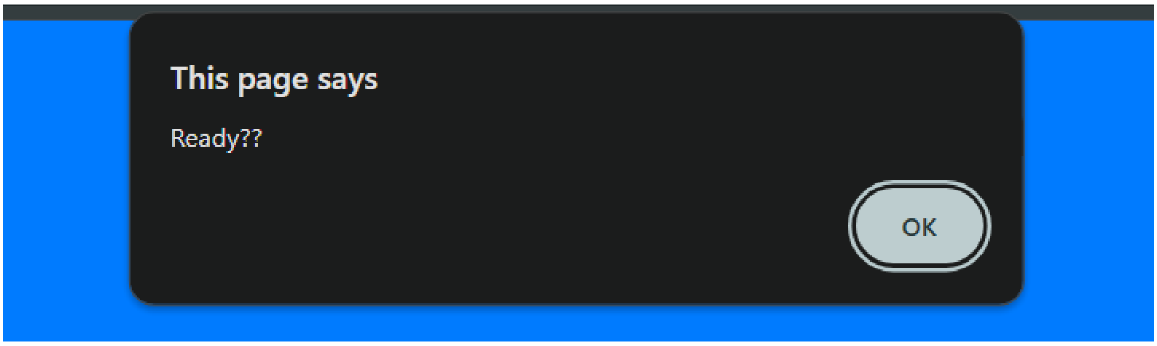



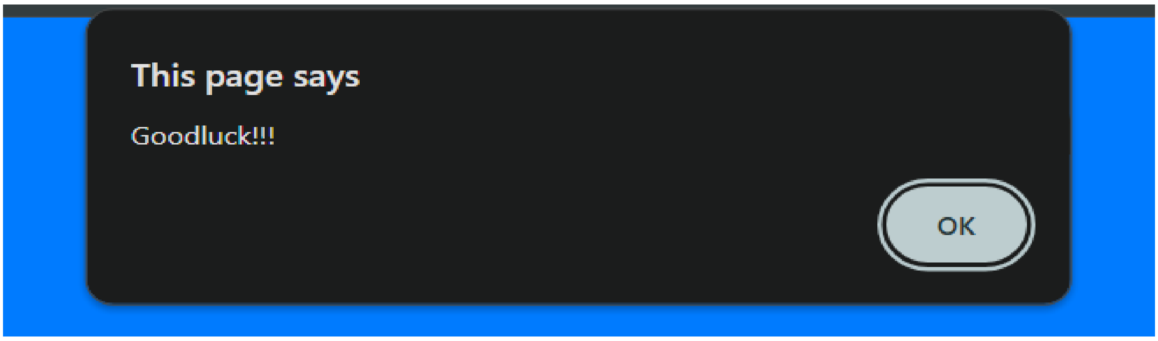

Start Quiz

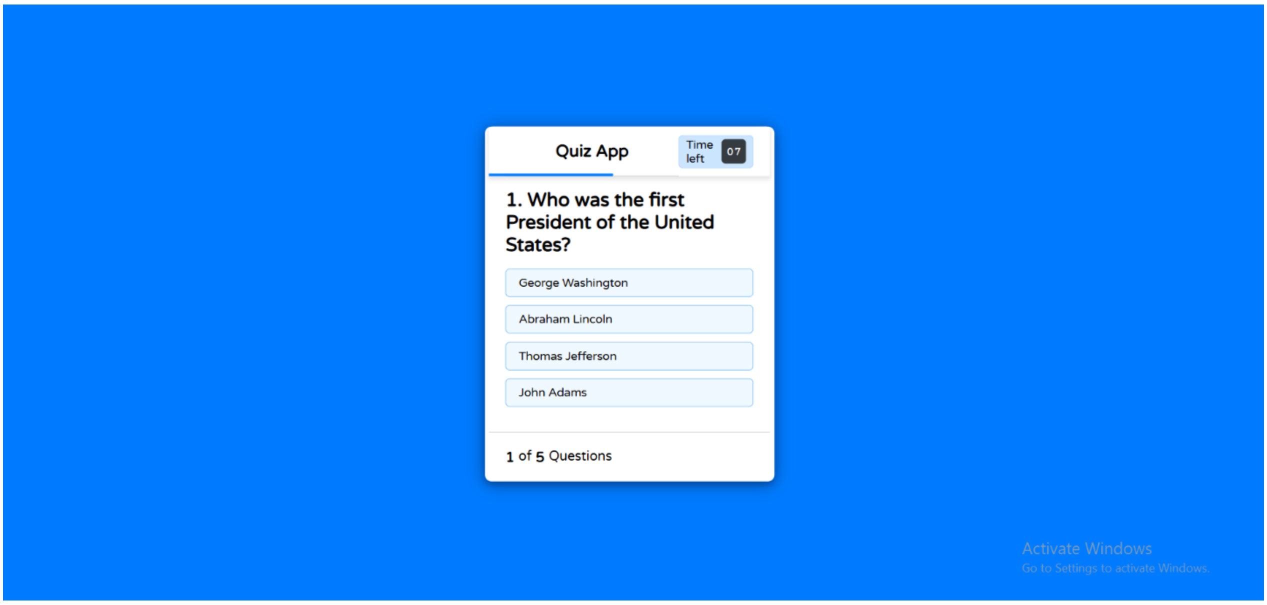

Select a Subject
Math
Science
History
Geography
Literature
General Knowledge
Activate Windows
Go to Settings to activate Windows.

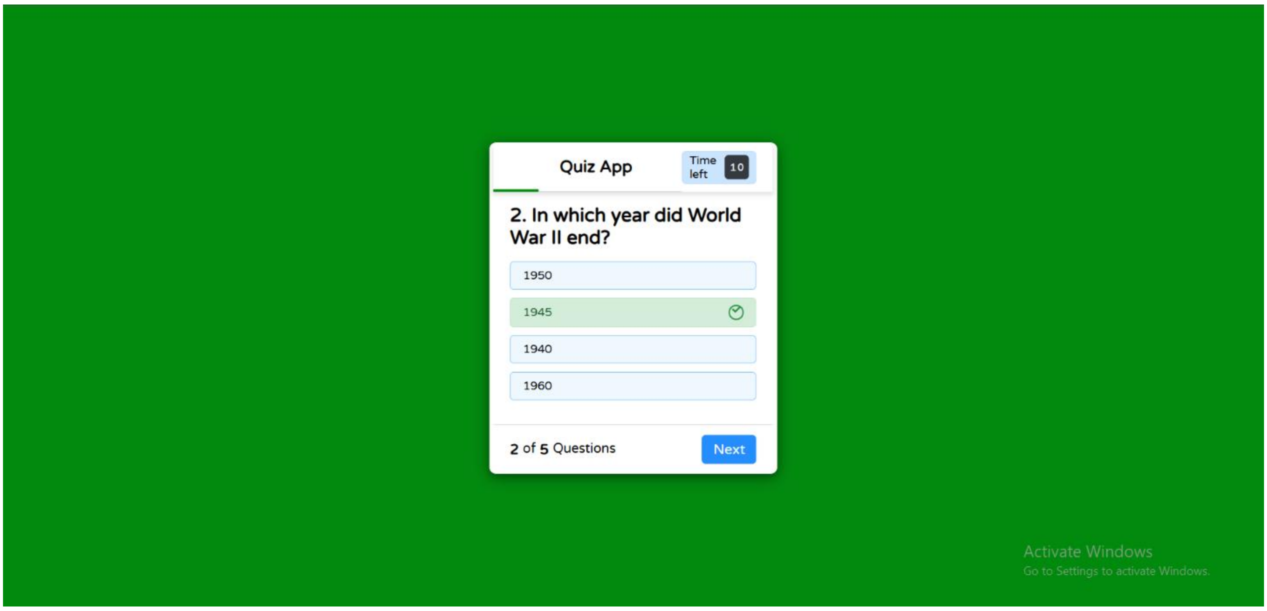

Select Difficulty
Easy
Medium
Hard
Activate Windows
Go to Settings to activate Windows.

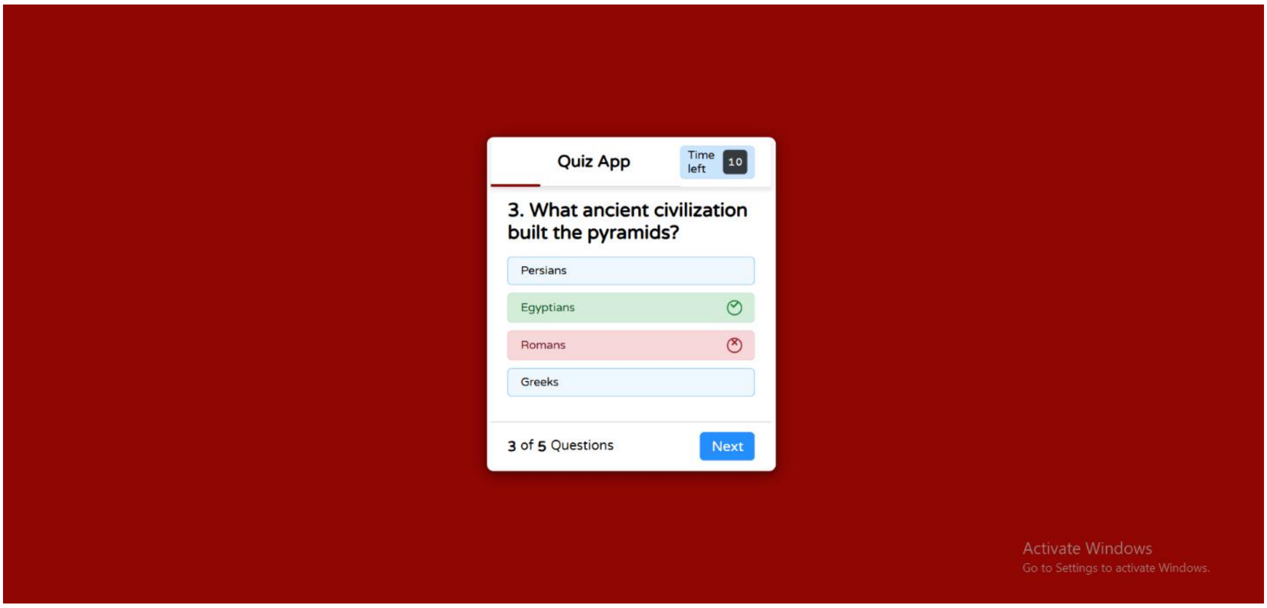

Rules of this quiz
* You only have 10 seconds for each question.
* You have to select an option before you can go to the next question.
* There's no going back after selecting an option.
* You cannot select an option once the timer goes off.
* You cannot exit the quiz while playing.
Exit Quiz
Continue
Activate Windows
Go to Settings to activate Windows.

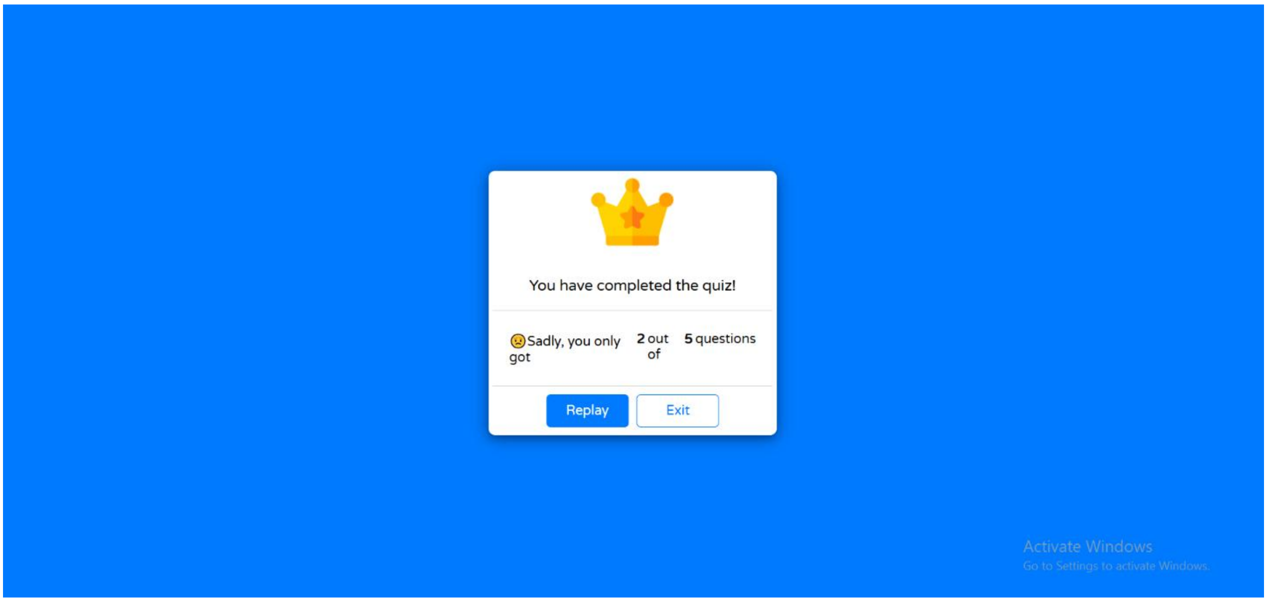

This page says
Ready??
OK


This page says
Goodluck!!!
OK


Quiz App
Time left 07
1. Who was the first President of the United States?
George Washington
Abraham Lincoln
Thomas Jefferson
John Adams
1 of 5 Questions
Activate Windows
Go to Settings to activate Windows.


Quiz App
Time left 10
2. In which year did World War II end?
1950
1945
1940
1960
2 of 5 Questions
Next
Activate Windows
Go to Settings to activate Windows.


Quiz App
Time left 10
3. What ancient civilization built the pyramids?
Persians
Egyptians
Romans
Greeks
3 of 5 Questions
Next
Activate Windows
Go to Settings to activate Windows.




## Conclusion

In conclusion, the Online Quiz System exemplifies the perfect blend of core Java programming concepts and interactive software design. By utilizing principles such as object-oriented programming, exception handling, and dynamic user interfaces, the application offers a robust, scalable, and engaging platform. The inclusion of features like secure authentication, real-time quiz generation, and progress tracking ensures users are equipped with everything they need to succeed. Additionally, principles such as loose coupling, high cohesion, and exception handling contribute to the system's efficiency and maintainability, ensuring it remains adaptable to future enhancements. This project not only highlights the power of Java but also sets a foundation for future, more complex applications, making it a versatile tool in both educational and programming contexts.